

IoTrix 2.0 – Technical Progress Report

Intelligent Landslide Detection & Early Warning System

Track: IoT Product & System Architecture Engineering

1. Core Overview

IoTrix 2.0 provides continuous landslide risk monitoring in off-grid environments using distributed ESP32 edge nodes with local anomaly detection, LoRaWAN mesh communication, and a Raspberry Pi gateway for localized slope-risk evaluation and cloud sync.

2. Architecture & Data Pipeline

- **ESP32 Edge Nodes:** Sample soil moisture, tilt, rainfall, and vibration; execute lightweight local ML anomaly models.
- **LoRaWAN Uplink:** Low-power, long-range telemetry transmission to gateway.
- **Raspberry Pi Gateway:** Synthesizes multi-node telemetry, performs offline buffering, and evaluates localized slope risk.
- **Cloud & Dashboard:** MQTT ingestion, time-series storage, real-time UI, and automated emergency alert dispatch (GREEN → RED).

3. Current Progress & Validation

- **Software & Cloud:** Full microservices architecture, database schemas, and MQTT pipelines implemented.
- **ML & Gateway:** Prototype ML pipeline and slope-risk engine verified on Pi gateway architecture.
- **Testing:** Validated end-to-end data pipeline and risk escalation using Docker-based simulation framework.

4. Remaining Limitations & Next Steps

- **Hardware Assembly:** Physical ESP32 node deployment and custom protective enclosures pending.

- **Firmware & Calibration:** Need low-level C/C++ power optimization and lab calibration of physical sensors.
- **Field Validation:** On-site LoRaWAN propagation testing and model tuning against real-world geotechnical data required.

5. Technology Stack

Architecture Layer	Technologies & Tools
Hardware	ESP32, Raspberry Pi, Geotechnical Sensors
Communication	LoRaWAN, MQTT, 4G/LTE, Satellite Fallback
Edge Storage & ML	SQLite, TensorFlow Lite Micro / Lightweight ML
Cloud Infrastructure	GCP, TimescaleDB, FastAPI, WebSockets
Frontend UI	React, Tailwind CSS, Leaflet Maps
DevOps & Tooling	Docker, Terraform, Kubernetes